

MAT 017B Midterm 1

August 16, 2018

Name: Key

SID: _____

$$\begin{aligned} \int x^n dx &= \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 & \int \frac{1}{x} dx &= \ln |x| + C & \int e^x dx &= e^x + C \\ \int \sin x dx &= -\cos x + C & \int \cos x dx &= \sin x + C & \int \sec^2 x dx &= \tan x + C \\ \int \frac{1}{x^2 + a^2} dx &= \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C & \int_a^b f(x) dx &= F(b) - F(a) \\ \int_a^b f(x) dx &\approx \sum_{k=1}^n f(x_k) \Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x \\ \frac{d}{dx} \int_a^x f(t) dt &= f(x) & \frac{d}{dx} \int_{h(x)}^{g(x)} f(t) dt &= f[g(x)]g'(x) - f[h(x)]h'(x) \\ \int f'(g(x))g'(x) dx &= f(g(x)) + C & \int u dv &= uv - \int v du \end{aligned}$$